

Day 3: Operators

- Arithmetic operators
- Relational operators
- Logical operators
- Bitwise operators
- Assignment operators

Programs:

- Calculator
- Even/odd
- Largest of 3 numbers

First Understand This

Manam variables create chesam.

```
int a = 10;
```

```
int b = 20;
```

Now Java ki ela cheppali

- Add cheyyi
- Subtract cheyyi
- Compare cheyyi
- Check condition
- Store result

Ee panulu cheyyadaniki use ayye symbols ni **Operators** antaru.

Simple ga

Operator = Action cheppe symbol

Example

+

-

*

/
==
>
<
&&
||
=

Real Life Example

Suppose

You = 500 Rs

Friend = 300 Rs

Questions

Total money?

Difference?

Who has more?

Both have money?

Java lo ivanni operators tho chestham.

Types of Operators

Java lo chala operators unnayi.

Mana Day 3 syllabus lo

1. Arithmetic Operators
2. Relational Operators
3. Logical Operators
4. Bitwise Operators
5. Assignment Operators

Let's learn one by one.

1. Arithmetic Operators

Arithmetic ante Mathematics.

School lo use chesina operations.

Addition

Subtraction

Multiplication

Division

Modulus

Java Symbols

Operator Meaning

+	Addition
-	Subtraction
*	Multiplication
/	Division
%	Remainder

Addition (+)

```
int a = 10;
```

```
int b = 20;
```

```
System.out.println(a + b);
```

Output

30

Java internally

$10 + 20 = 30$

Subtraction (-)

```
System.out.println(20 - 10);
```

Output

10

Multiplication (*)

```
System.out.println(5 * 6);
```

Output

30

Division (/)

```
System.out.println(20 / 5);
```

Output

4

Important Concept

```
System.out.println(5 / 2);
```

Output

2

Students confuse avtaru.

Why not 2.5?

Reason:

5

--

2

Answer

2.5

But

5 and 2 are int

Java

Decimal remove chestundi

Output

2

If

`System.out.println(5.0 / 2);`

Output

2.5

Because

5.0 is double

Modulus Operator (%)

Most Important

Symbol

%

Returns

Remainder

Example

$10 \% 2$

Division

$10 \div 2$

Quotient = 5

Remainder = 0

Output

0

Example

$11 \% 2$

$11 \div 2$

Quotient =5

Remainder =1

Output

1

Example

$17 \% 5$

$17 \div 5$

Quotient=3

$3 \times 5 = 15$

$17 - 15 = 2$

Output

2

Why % is Important?

Almost every interview lo use chestaru.

Example

Even Odd

Leap Year

Prime Number

Palindrome

Armstrong

Fibonacci

Arithmetic Program

```
public class ArithmeticDemo
{
    public static void main(String args[])
    {
        int a=20;
        int b=10;

        System.out.println("Addition = "+(a+b));
        System.out.println("Subtraction = "+(a-b));
        System.out.println("Multiplication = "+(a*b));
        System.out.println("Division = "+(a/b));
        System.out.println("Remainder = "+(a%b));
    }
}
```

Output

Addition =30

Subtraction=10

Multiplication=200

Division=2

Remainder=0

2. Relational Operators

Relation ante compare.

Question adugutundi.

Equal?

Greater?

Smaller?

Always returns

true

or

false

Operators

Operator Meaning

>	Greater than
<	Less than
>=	Greater or Equal
<=	Less or Equal
==	Equal
!=	Not Equal

Example

```
int a=10;
```

```
int b=20;
```

```
System.out.println(a>b);
```

Question

10 >20?

Answer

false

Example

```
System.out.println(20>10);
```

Output

true

Example

```
System.out.println(10==10);
```

Output

true

Example

```
System.out.println(10!=20);
```

Question

10 not equal 20?

Output

true

Very Important Difference

Many students mistake:

=

and

==

=

Assignment

Store value

Example

int a=10;

==

Comparison

10==10

Output

true

3. Logical Operators

Suppose

College rules

Attendance >75%

AND

Fees Paid

Only then exam.

One condition saripodu.

Multiple conditions.

Logical operators use chestham.

Operators

Operator Meaning

&& AND

|| OR

! NOT

AND (&&)

Rule

Both conditions true

Then only true

Truth Table

Condition1	Condition2	Result
------------	------------	--------

T	T	T
---	---	---

T	F	F
---	---	---

F	T	F
---	---	---

F	F	F
---	---	---

Example

10>5 && 20>15

true && true

Output

true

Example

10>20 && 5<9

false && true

Output

false

OR (||)

Rule

At least one true.

Truth Table

Condition1	Condition2	Result
------------	------------	--------

T	T	T
---	---	---

Condition1	Condition2	Result
------------	------------	--------

T	F	T
---	---	---

F	T	T
---	---	---

F	F	F
---	---	---

Example

10>20 || 5<9

false || true

Output

true

NOT (!)

Reverse

true becomes false

false becomes true

Example

!(10>5)

!(true)

false

Where Logical Operators are Used?

Login

ATM

Voting

Exam Eligibility

Age Validation

Password Checking

Bank

College Admission

Short-Circuit Evaluation (Very Important Interview Topic)

&& and || Java lo **short-circuit** chestayi.

Example:

```
int a = 5;
```

```
int b = 0;
```

```
if (a > 10 && (10 / b > 1)) {  
    System.out.println("Hello");  
}
```

Java first condition check chestundi:

$a > 10 \rightarrow \text{false}$

Already && lo first false ayindi kabatti second condition check cheyyadu. So 10/b execute avvadu. Program safe ga run avutundi.

Alage || lo first condition true ayite second condition check cheyyadu.

4. Bitwise Operators

Idi beginners ki konchem new topic.

Computer lo data **0 and 1 (bits)** form lo untundi.

5 → 0101

3 → 0011

Bitwise operators bits meeda work chestayi.

Operator Meaning

& Bitwise AND

| Bitwise OR

^ Bitwise XOR

~ Bitwise NOT

Decimal to Binary

5 = 0101

3 = 0011

Bitwise AND (&)

Rule

1 & 1 = 1

Else 0

0101

0011

0001

Answer

1

Java

```
System.out.println(5 & 3);
```

Output

1

Bitwise OR (|)

Rule

Any one 1 ayina output 1.

0101

0011

0111

Output

7

Bitwise XOR (^)

Rule

Same bits → 0

Different bits → 1

0101

0011

0110

Output

6

Bitwise NOT (~)

Bits ni reverse chestundi.

00000101

↓

11111010

Java two's complement use chestundi, kabatti result negative vastundi.

System.out.println(~5);

Output:

-6

Remember:

$\sim n = -(n + 1)$

Example:

$\sim 10 = -11$

$\sim 20 = -21$

5. Assignment Operators

Assignment ante value ni variable lo store cheyyadam.

Basic

`int a=10;`

=

`a=5;`

Store 5

+=

`a+=5;`

Meaning

`a=a+5;`

Example

`a=10;`

`a+=5;`

10+5

Output

15

-=

a-=2;

Means

a=a-2;

***=**

a*=3;

Means

a=a*3;

/=

a/=2;

Means

a=a/2;

%=

a%=2;

Means

a=a%2;

Assignment Operator Program

```
public class AssignmentDemo {  
    public static void main(String[] args) {
```

```
        int a = 10;
```

```
        a += 5;
```

```
        System.out.println(a);
```

```
a -= 2;
System.out.println(a);

a *= 3;
System.out.println(a);

a /= 2;
System.out.println(a);

a %= 4;
System.out.println(a);
}
}
```

Output

```
15
13
39
19
3
```

Program 1: Calculator

```
import java.util.Scanner;

public class Calculator {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
```

```
System.out.print("Enter first number: ");
int a = sc.nextInt();

System.out.print("Enter second number: ");
int b = sc.nextInt();

System.out.println("Addition = " + (a + b));
System.out.println("Subtraction = " + (a - b));
System.out.println("Multiplication = " + (a * b));

if (b != 0) {
    System.out.println("Division = " + (a / b));
    System.out.println("Remainder = " + (a % b));
} else {
    System.out.println("Division and Modulus by zero are not allowed.");
}

sc.close();
}
```

Sample Output

Enter first number: 20

Enter second number: 5

Addition = 25

Subtraction = 15

Multiplication = 100

Division = 4

Remainder = 0

Program 2: Even or Odd

Logic:

- $\text{Number \% 2} == 0 \rightarrow \text{Even}$
- $\text{Number \% 2} != 0 \rightarrow \text{Odd}$

```
import java.util.Scanner;
```

```
public class EvenOdd {  
    public static void main(String[] args) {  
  
        Scanner sc = new Scanner(System.in);  
  
        System.out.print("Enter a number: ");  
        int num = sc.nextInt();  
  
        if (num % 2 == 0) {  
            System.out.println(num + " is Even");  
        } else {  
            System.out.println(num + " is Odd");  
        }  
  
        sc.close();  
    }  
}
```

Sample Output

```
Enter a number: 17
```

```
17 is Odd
```

Program 3: Largest of 3 Numbers

Logic:

- Compare a, b, and c.
- Biggest value print cheyyali.

```
import java.util.Scanner;
```

```
public class LargestOfThree {  
    public static void main(String[] args) {  
  
        Scanner sc = new Scanner(System.in);  
  
        System.out.print("Enter first number: ");  
        int a = sc.nextInt();  
  
        System.out.print("Enter second number: ");  
        int b = sc.nextInt();  
  
        System.out.print("Enter third number: ");  
        int c = sc.nextInt();  
  
        if (a >= b && a >= c) {  
            System.out.println(a + " is the largest");  
        } else if (b >= a && b >= c) {  
            System.out.println(b + " is the largest");  
        } else {  
            System.out.println(c + " is the largest");  
        }  
  
        sc.close();  
    }  
}
```

```
}  
}
```

Sample Output

Enter first number: 25

Enter second number: 18

Enter third number: 42

42 is the largest

Interview Questions (Must Know)

1. Operator ante enti?
2. Arithmetic operators enti?
3. / mariyu % madhya difference enti?
4. = mariyu == difference enti?
5. && mariyu || difference enti?
6. Short-circuit evaluation ante enti?
7. 5 / 2 output enti? Enduku?
8. 5 % 2 output enti?
9. Bitwise & mariyu Logical && difference enti?
10. ~5 output enti? Enduku?
11. a += 5 equivalent statement enti?
12. Even/Odd program lo % enduku use chestham?
13. Largest of 3 numbers program logic explain cheyyandi.